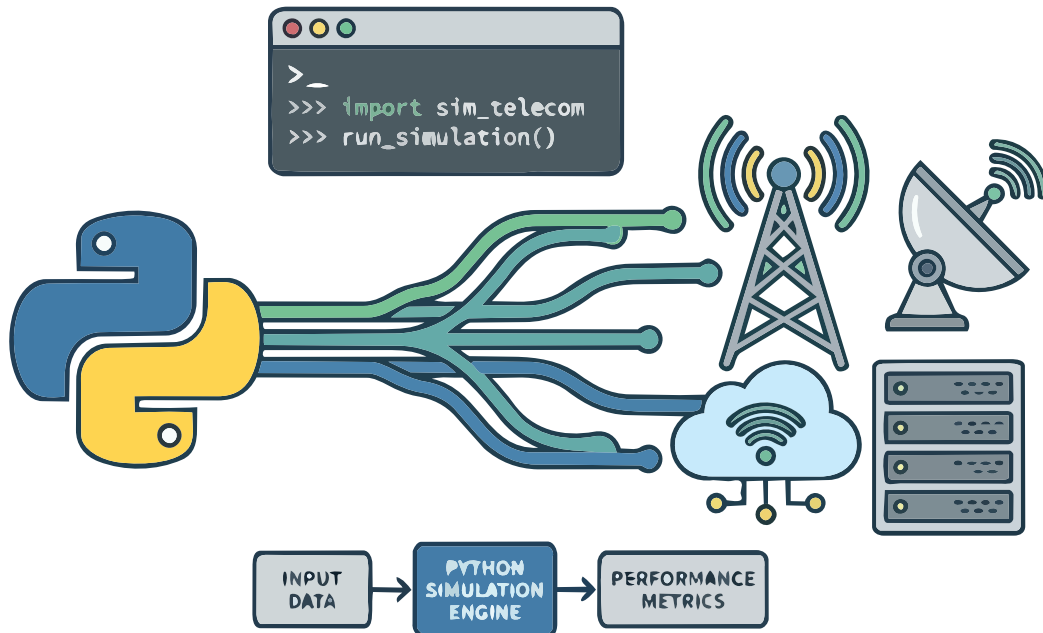


Simulating Telecom Applications with Python

Did you know that apart from its many other utilities, Python can also be used for simulating a range of telecom and wireless applications?



Several programming libraries are being used today for simulating and deploying real world applications in telecommunications, software defined radio (SDR), antenna design, electromagnetic pattern analysis, satellite communications, etc. The Indian telecom and wireless market is currently investing in 6G, silicon carbide (SiC), and gallium nitride (GaN) technologies, and other future wireless domains.

A number of programming platforms and languages are being used for radio frequency analysis and antenna design for real world telecom and wireless applications:

- **EM (electromagnetic) simulation tools** (CST, HFSS, FEKO): Core for antenna design
 - **Circuit tools** (ADS, LTSpice): RFID chip and front-end design
 - **Programming tools** (Python, MATLAB): Algorithm and signal modelling
 - **PCB tools** (KiCad, Altium, EasyEDA): Physical implementation
 - **System tools** (Simulink, GNU Radio): Full RFID communication workflow
- Python, GNU Radio, Simulink, Octave, and Julia are

among the programming languages being used for signal processing and RF modelling, RF analysis and automation, RFID reader prototyping, communication system simulation, and high-speed scientific computing so that real-world applications can be analysed before actual deployment on site.

Python is a popular programming language for various domains including blockchain, cloud computing, cybersecurity, digital forensics, image processing, etc. It is now also available for radio frequency (RF), software defined radio (SDR), antenna design and many other telecom and wireless applications (Table 2).

OpenEMS: Free and open source platform for electromagnetic analysis, radio and antenna engineering

<https://www.openems.de>

OpenEMS is available under free distribution as an electromagnetic field solver for telecom, radio and antenna engineering applications. Its programming interfaces are available for Python, Octave, and MATLAB for simulation by researchers and engineers.